A Comparative Analysis of Dysarthric and Non-Dysarthric Speech Features using the TORGO Dataset

Shrinivas Sampath Muthupalaniyappan, Aarefa Zuzer Bhurka, Matej Malesevic

sm3533@rit.edu, ab4911@rit.edu, mm7896@rit.edu

Abstract

The aim of this research is to explore the potential use of the TORGO database for individuals with speech impairments through three experiments. Presently, publicly available speech datasets and ASR models are mainly geared towards non-dysarthric speech, which creates difficulties in developing effective Automatic Speech Recognition (ASR) systems for people with speech disabilities. Popular ASR systems like Amazon Alexa, Apple Siri, Google Home, Microsoft Cortona, and IBM Watson have been found to have high word error rates ranging from 20% to 70% for individuals with spastic and dysarthria speech. To investigate dysarthric speech using the TORGO database, the study presents three research questions.

RQ1. What is the Word Error Rate of current ASR models such as Wave2Vec 2.0 when tested on dysarthria speech?

RQ2. How similar are dysarthric speakers to other dysarthric speakers and non-dysarthric speakers?

RQ3. Which key speech features can effectively differentiate dysarthria subtypes, and how can their integration enhance ASR performance for dysarthric speakers?

Index Terms: speech recognition, dysarthria, augmentative and alternative communication

1. Introduction

Dysarthria is a neurological motor speech disorder that arises from damage or injury to the nervous system, which affects the muscles responsible for speech production [1]. The resulting speech impairments may manifest in various ways, including articulatory imprecision, slow speaking rate, reduced phonetic contrasts, and abnormal prosody. Dysarthria can be caused by various neurological conditions, such as stroke, traumatic brain injury, Parkinson's disease, multiple sclerosis, and amyotrophic lateral sclerosis (ALS) [1].

Speech production involves a complex interplay of various articulators, including the tongue, lips, and jaw, as well as the coordination of respiratory, phonatory, and articulatory systems [2]. A deeper understanding of the speech features in dysarthric speakers, such as tongue position, phones, and other relevant features, is essential for advancing the field of automatic speech recognition (ASR) and developing more effective ASR systems tailored to the needs of individuals with dysarthria.

Comparing the speech features of dysarthric speakers to those of non-dysarthric speakers can help to identify the unique characteristics of dysarthric speech and provide a basis for improving ASR system performance in recognizing dysarthric speech [**?**]. Additionally, investigating the similarities between dysarthric speakers can contribute to a more comprehensive understanding of the commonalities and variations within dysarthric speech patterns, which may lead to more targeted and effective ASR systems for individuals with different underlying neurological conditions.

The TORGO dataset [3] is a valuable resource for exploring and comparing the speech features of dysarthric and non-dysarthric speakers, as it contains a diverse range of audio recordings and metadata from individuals with and without dysarthria. By analyzing and comparing these speech features using cosine similarity [4], this study aims to provide insights into the speech patterns of dysarthric speakers and contribute to the development of improved ASR systems for individuals affected by this motor speech disorder.

The primary objectives of this study involve a multifaceted approach to understanding and comparing the speech features of dysarthric and non-dysarthric speakers. First, the study aims to test current ASR models such as Wave2vec 2.0, which are trained only on non-dysarthric data from 960 hours of the Librispeech corpus [5], against speech-impaired data from the TORGO database [3]. Then, for our second research question, we will use Praat [6], a speech analysis tool, to extract features and compare cosine similarity to assess the differences and similarities between the speech patterns of dysarthric and non-dysarthric speakers. Next, for our third research question, We investigate relationships among various dysarthria types and aim to identify key speech features to improve ASR performance. Using data from the TORGO database and our own dataset, we analyze and visualize these features to better understand their potential impact on ASR systems for dysarthric speakers.

2. Related work

2.1. Dysarthria and Speech Features

Dysarthria is characterized by impairments in the production of speech due to damage or injury to the nervous system [1]. It affects the muscles responsible for speech production, leading to various speech impairments such as articulatory imprecision, slow speaking rate, reduced phonetic contrasts, and abnormal prosody. The specific speech features affected by dysarthria may vary depending on the underlying neurological condition, such as stroke, traumatic brain injury, Parkinson's disease, multiple sclerosis, or amyotrophic lateral sclerosis (ALS) [1].

Previous studies have investigated the speech features associated with different types of dysarthria, with particular attention paid to articulatory aspects such as tongue position, phones, and other relevant features [7]. For example, studies have shown that individuals with dysarthria may have difficulties with specific phonemes, such as fricatives and affricates, due to reduced articulatory precision [8].

2.2. Automatic Speech Recognition for Dysarthric Speakers

Automatic speech recognition (ASR) systems have shown great promise for facilitating communication and improving the quality of life for individuals with speech impairments [9]. However, the performance of conventional ASR systems can be significantly degraded when confronted with dysarthric speech, due to the unique characteristics of dysarthric speech and the limited amount of training data available for this population [**?**].

Several studies have attempted to address the challenges associated with ASR for dysarthric speakers, such as adapting existing ASR systems to better accommodate the speech features of dysarthric individuals [10] or developing new ASR systems specifically designed for dysarthric speech [11] [12]. These studies have demonstrated the potential for improving ASR performance for dysarthric speakers by incorporating knowledge of dysarthric speech features and patterns.

2.3. Comparing Dysarthric and Non-Dysarthric Speech

Previous research comparing the speech features of dysarthric and non-dysarthric speakers has provided valuable insights into the unique characteristics of dysarthric speech [13]. For example, studies have reported that dysarthric speakers may exhibit reduced articulatory precision, slower speaking rates, and less distinct phonetic contrasts compared to non-dysarthric speakers [13].

Such comparative analyses can help identify the specific speech features that are affected by dysarthria, providing a foundation for the development of more effective speech therapy methods and assistive technologies, including ASR systems tailored for dysarthric speakers [14]. Furthermore, examining the similarities between dysarthric speakers can shed light on the commonalities and variations within dysarthric speech patterns, which may lead to more targeted and effective treatments and technologies for individuals with different types of dysarthria [15].

3. Dataset

3.1. TORGO

The TORGO dataset [3] was used in this study to analyze the speech features of dysarthric and non-dysarthric speakers. The dataset consists of audio recordings and corresponding transcriptions of dysarthric speakers affected by different neurological conditions, as well as age- and gender-matched non-dysarthric speakers. The dataset includes a diverse range of speakers with various severity levels of dysarthria, providing a comprehensive resource for the analysis of speech features in dysarthric and non-dysarthric speakers. For each speaker the TORGO database contains audio data recorded in two different mics, the PHN folder contains the annotated files for the equivalent audio files, and a txt folder contains the audio translation in the English language. To facilitate a comprehensive comparison between dysarthric and non-dysarthric speech, our case study involves the utilization of only four male speakers, two with dysarthria and two without. The count of male dysarthric speakers accounted for 179 and the count of male non-dysarthric speakers accounted for 178. The distribution of the dataset is provided in Table 1.

Dataset	Count
Male TORGO non-dysarthric	178
Male TORGO dysarthric	179
Male dysarthric recordings	30

Table 1: *Dataset*

S.No	One-word Transcript	Two-word Transcript
1.	Hello	Thank you
2.	Yes	Excuse me
3.	No	Goodbye now
4.	Stop	Need help
5.	Go	Happy birthday
6.	Help	I'm sorry
7.	Water	Nice day
8.	Food	Bad pain
9.	Pain	Cold outside
10.	Sleep	Hot today
11.	Bath	Rain coming
12.	Up	Quiet please
13.	Down	Loud noise
14.	Love	Slow down
15.	Hate	Fast car

Table 2: *Transcripts*

3.2. Recordings

As a part of our analysis, we incorporated dysarthric speech data from one of our male team members. We recorded the data in a quiet environment using a standard mobile phone and compiled a transcript of 30 commonly used one-word and two-word phrases (refer to Table 2). Combining this with the TORGO database, we had a dataset of 387 speech samples for our analysis (refer to Table 1).

4. Methodology

In this section, we propose our approach for all three research questions. Figure 1 displays the methodology used to address RQ1. We will be going through a detailed explanation for each of the code blocks of Figure 1 in Section 4.1. The approach for research question 2 will be explained in Section 4.2. Similarly, research question 3 will be explained in Section 4.3

4.1. RQ1 Methodology

In this method, we will be testing the current Automatic Speech Recognition (ASR) model such as Wave2Vec 2.0 against the dysarthric data. Wave2vec 2.0 is a speech recognition model developed by Facebook AI research scientists. In our RQ1 methodology, we will be using the Wave2vec 2.0 model trained on 960 hours of unlabeled Librispeech corpus. The main reason behind using the 960 hours Librispeech model is because it was trained with non-dysarthric data in the English language.

For this methodology, we will be using the TORGO database[3] as our input. Similar audio files for non-dysarthric and dysarthric speakers from TORGO are passed one by one to the Wave2Vec 2.0 model to get the generated transcript.

In the next step, we will explain the Figure 4.1 block dia-

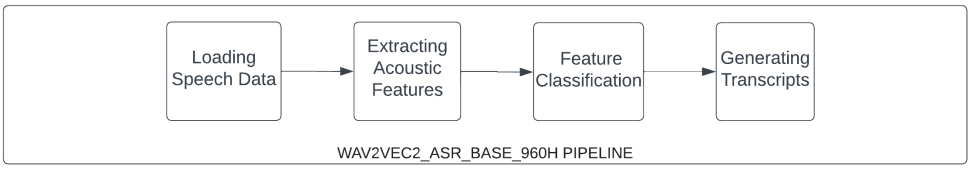


Figure 1: *RQ1 Approach*

gram. In order for the Wave2vec 2.0 model to work we have to create a pipeline with four necessary steps i) Loading Speech data ii) Extracting Acoustic Features iii) Feature classification iv) Generating Transcripts as shown in Figure 4.1.

Using the torchaudio library, a wave2vec 2.0 model pipeline is created for 960-hour speech data. Then a single audio file for a non-dysarthric sample is loaded in the pipeline. The TORGO audio was recorded on a 16khz sample rate. A torch audio model checks if the 16khz sample rate is present in the Wav2vec 2.0 model. If the loaded sample rate is present then the wave2vec loads the audio file for the next step. Using the loaded model we will call a function to extract the features. First, 12 Mel Cepstral Coefficient (MFCC) is extracted from the loaded audio file. Then feature classification is performed on the extracted features. A probability will be generated on the final softmax layer containing 32 alphabet nodes for the entire audio file. Then for the generated probability, we will use a context-independent decoder to convert the sequence probability into transcripts. Then we will repeat the same process for the dysarthric audio data as well.

In the end, we calculate the word error rate (refer equation 1) of the converted transcript as compared to the original transcript. This measure of word error rate will provide insights into the performance of current ASR models against dysarthric speech, which are typically trained on non-dysarthric data.

$$\textbf{WER} = \frac{S + I + D}{N} \tag{1}$$

4.2. RQ2 Methodology

In this section, we will talk about our approach to solving research question two. Figure 2 shows the approach for the RQ2.

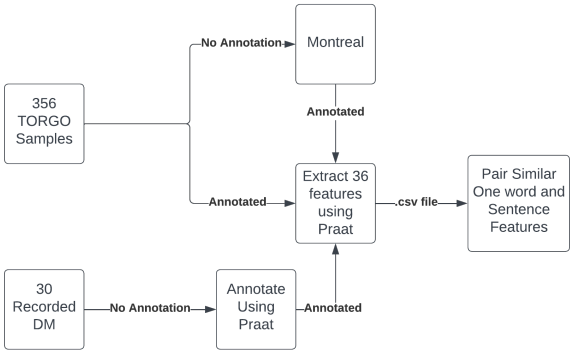


Figure 2: *RQ2 Approach*

The methodology for RQ2 aims to identify whether there are any feature similarities among non-dysarthric speakers, dysarthric speakers, or between non-dysarthric and dysarthric speakers.

Before analyzing the features, we must annotate and extract the features using praat. Firstly, The TORGO database has 356 samples containing male non-dysarthria and dysarthria speakers. Even though more than 90% of the database has annotated files for non-dysarthria speakers attached to the respective speaker. 10% of the data does not have annotation. Annotation here refers to transcribing the .wav audio files to the textgrid format in ARPABET style. For the remaining 10% of audio files, we are using a forced alignment tool called Montreal to transcribe into ARPABET.

Montreal already has an English acoustic model and a pronunciation dictionary to convert the .wav audio files to the ARPABET. Montreal requires .txt files containing the actual English transcript and the equivalent audio file to convert into textgrid format with ARPABET transcription on it. The input for the Montreal should be stored in the same directory containing the same audio file and the .txt transcript containing the actual transcription of the audio file. Then we have to run the Montreal align command containing the directory to the input file, English acoustic model, English pronunciation dictionary, and output directory. If every directory is properly given the output textgrid files will be stored in the output directory. Refer to Figure 3 for annotated data by Montreal.

Secondly, we recorded 30 speech samples from a team member with dysarthria, who had a good understanding of speech processing. The annotations for the transcriptions were also carried out by the same team member to ensure the most accurate results, as they possessed expertise in both the recorded data and the transcription process.

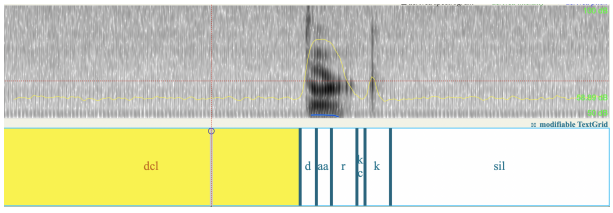


Figure 3: *Montreal Transcription*

Finally, all the textgrid files, .txt files, and the .wav files for both the TORGO and the recorded dysarthric male data were stored in the same directory to extract the relevant features using the Praat script. A praat script was written to extract the 36 features shown in Table 3. For each audio file depending on the number of consonants and vowel boundaries, the count of the consonant and vowel was calculated, and then the summed total for each of the 36 features was divided by the respective

S.No	Features	S.No	Features
1.	conso_duration_avg	19.	f0Point3_vow_avg
2.	vow_duration_avg	20.	f0Point4_vow_avg
3.	pitchmax_conso_avg	21.	f0Point5_vow_avg
4.	pitchmin_conso_avg	22.	f0Point6_vow_avg
5.	pitchmax_vow_avg	23.	f0Point7_vow_avg
6.	pitchmin_vow_avg	24.	f0Point8_vow_avg
7.	f0Point1_conso_avg	25.	f0Point9_vow_avg
8.	f0Point2_conso_avg	26.	f0Point10_vow_avg
9.	f0Point3_conso_avg	27.	max_dB_conso_avg
10.	f0Point4_conso_avg	28.	min_dB_conso_avg
11.	f0Point5_conso_avg	29.	mean_dB_conso_avg
12.	f0Point6_conso_avg	30.	max_dB_vow_avg
13.	f0Point7_conso_avg	31.	min_dB_vow_avg
14.	f0Point8_conso_avg	32.	mean_dB_vow_avg
15.	f0Point9_conso_avg	33.	jitter
16.	f0Point10_conso_avg	34.	shimmer
17.	f0Point1_vow_avg	35.	voicebreaks
18.	f0Point2_vow_avg	36.	degree_of_voice_breaks

Table 3: *Extracted Features*

consonant and vowel count. The consonant and vowel level features are extracted to differentiate between dysarthric and non-dysarthric speakers. The extracted features carry high-level information about the speaker's consonant and vowel information.

At last cosine similarity between the extracted features is calculated for similar words between three dysarthria speakers and two non-dysarthria speakers in all possible combinations.

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|} \tag{2}$$

4.3. RQ3 Methodology

To explore the connections between various dysarthria types and pinpoint crucial speech features for enhancing ASR performance, we first gather a collection of speech recordings from two male speakers with dysarthria available in the TORGO database, as well as one additional male speaker from our own generated dataset. After preparing the datasets, we calculate a wide range of speech features, which includes the 13 top-ranked features from our study as shown in the table **??**, and visualize them using techniques like t-SNE to reduce dimensionality. This helps us identify trends, groupings, and relationships among the diverse dysarthria types based on their speech characteristics.

Next, we use feature selection methods to determine the most important features for distinguishing between dysarthria types. We apply Recursive Feature Elimination (RFE) for feature selection, ranking the speech features in terms of their importance for distinguishing between dysarthria types. Figure 4 shows the speech feature distribution that distinguished each speaker. We compile a list of the top-ranked speech features that could potentially improve ASR performance for dysarthric speakers. This list will serve as a valuable resource for researchers and engineers working on ASR systems tailored to dysarthric speech.

Lastly, we analyze the effectiveness of these top-ranked features in differentiating dysarthria types and discuss their potential impact on ASR performance. By identifying the most relevant speech features, our study aims to contribute to the ongoing efforts to improve ASR systems for individuals with dysarthria.

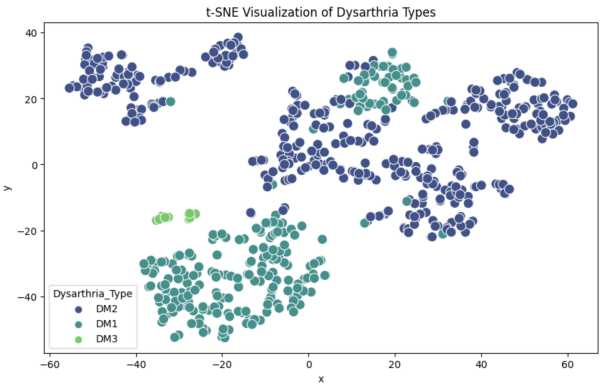


Figure 4: *Dysarthric Speaker Groups - DM1, DM2, and DM3 referes to Dysarthric Male 1, Dysarthric Male 2 and Dysarthric Male 3 respectively.*

Speech Features	**Rank**
ZCR	1
MFCC	2
Formant	3
Spectral Rolloff	4
Spectral Centroid	5
MFCC Delta 1	6
MFCC Delta 2	7
energy	8
Spectral Flux	9
Chroma	10
LPC 2	11
Pitch	12
LPC 1	13

Table 4: *Ranked feature list - The lower the rank, the higher the importance of the feature.*

5. Discussion

The results of our research provide valuable insights into the challenges and opportunities associated with the development of ASR systems for individuals with speech impairments. Our investigation into the TORGO database's utility for dysarthric speech recognition revealed significant performance disparities between current ASR models and their ability to accurately recognize dysarthric speech.

5.1. Word Error Rate of Current ASR Models

In response to RQ1, our experiments revealed that current ASR models, such as Wave2Vec 2.0, exhibited high word error rates (WER) for dysarthric speech samples in the TORGO database (refer Table 5). This suggests that existing models are not adequately trained to handle the unique characteristics of dysarthric speech, and as a result, their performance is considerably lower for individuals with speech impairments. This highlights the need for further research and development in the area of ASR systems tailored to dysarthric speech recognition.

Utterance	Audio Type	Generated Transcript	WER
Don't ask me to carry an oily rag like that.	non-dysarthric	DON'T ASK ME TO CARRY AN OILY REGG LIKE THAT	9.1%
	dysarthric	IS DON AGUME DO GIY IN OEEGUEUBE	100%
Horn	non dysarthric	HORN	0
	dysarthric	OR	100%

Table 5: *RQ1 Result*

5.2. Speaker Similarity

Our second research question (RQ2) aimed to analyze the similarity between dysarthric speakers and non-dysarthric speakers. Our findings revealed that dysarthric speakers exhibited considerable variability in their speech characteristics, with some speakers being more similar to other dysarthric speakers in the word pair comparison (refer Figure 5), while non-dysarthric speakers shared greater similarities with dysarthric speakers in sentence pair comparison (refer Figure 6). This suggests that developing ASR systems for dysarthric speech may require the consideration of individual speaker characteristics and the adoption of personalized approaches to improve recognition performance.

Vector	'yes' | 'go'
<NDM1,NDM2>	0.14 | 0.19
<NDM1, DM1>	-0.06 | -0.08
<NDM1, DM2>	0.08 | -0.274
<NDM1, Matt>	-0.23 | -0.07
<NDM2, DM1>	-0.13 | -0.62
<NDM2, DM2>	0.03 | -0.50
<NDM2, Matt>	-0.25 | -0.07
<DM1,DM2>	0.48 | 0.48
<DM1,Matt>	0.28 | 0.28
<DM2,Matt>	0.03 | 0.03

Figure 5: *One-word comparison. NDM and DM refer to Non-dysarthric and dysarthric speakers. 1,2 and 3 refer to different respective speakers. Matt denotes the recorded speech*

5.3. Key Speech Features and ASR Performance Enhancement

In response to RQ3, our research identified several key speech features that effectively differentiated between dysarthria subtypes. These features included articulation, prosody, and phonation, among others. As shown in figure 7 choosing the right features can increase the similarity ratio between the speech features of different speakers and such features could enhance ASR systems' accuracy and usability for individuals with speech impairments.

Vector	Sentence
<NDM1,NDM2>	0.28
<NDM1, DM1>	0.55
<NDM1, DM2>	-0.23
<NDM2, DM1>	-0.18
<NDM2, DM2>	-0.37
<DM1,DM2>	-0.06

Figure 6: *Sentence comparison. NDM and DM refer to Non-dysarthric and dysarthric speakers. 1 and 2 refer to different respective speakers.*

6. Future Work

In light of our findings, future research should focus on several key areas to further advance the development of ASR systems for individuals with speech impairments. First, there is a need to develop dysarthria-specific ASR models, as current models exhibit high word error rates for dysarthric speech. Second, given the variability among dysarthric speakers, it is essential to explore the development of personalized ASR systems that can adapt to individual speaker characteristics, leading to improved recognition performance. Furthermore, expanding the TORGO database with more diverse samples from various dysarthria subtypes and severity levels would provide a more comprehensive training set, thereby enhancing the performance of ASR systems for dysarthric speech. Lastly, future research should investigate additional speech features that could be integrated into ASR models to improve performance for dysarthric speakers, such as speaker adaptation and context-aware processing. By addressing these challenges, we can move towards a more inclusive and accessible future for people with speech disabilities.

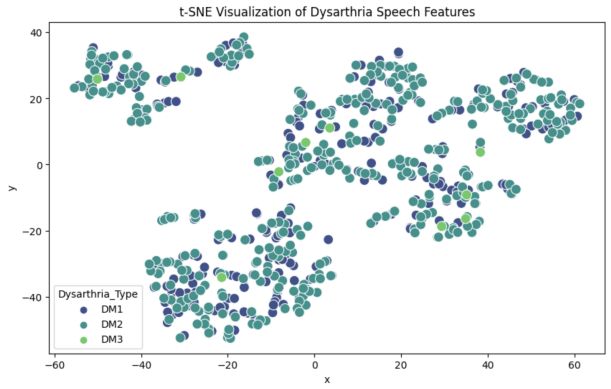


Figure 7: *Dysarthric Speaker MFCC Feature - 2-dimensional visualization of MFCC features of three groups of speakers. DM1, DM2, and DM3 refer to Dysarthric Male 1, Dysarthric Male 2, and Dysarthric Male 3 respectively.*

7. References

[1] P. Enderby, "Chapter 22 - disorders of communication: dysarthria," in *Neurological Rehabilitation*, ser. Handbook of Clinical Neurology, M. P. Barnes and D. C. Good, Eds. Elsevier, 2013, vol. 110, pp. 273–281. [Online]. Available: https://www.sciencedirect.com/science/article/pii/B9780444529015000228

[2] P. F. MacNeilage, "The frame/content theory of evolution of speech production," *Behavioral and Brain Sciences*, vol. 21, no. 4, p. 499–511, 1998.

[3] F. Rudzicz, A. K. Namasivayam, and T. Wolff, "The torgo database of acoustic and articulatory speech from speakers with dysarthria," *Language Resources and Evaluation*, vol. 46, pp. 523–541, 2012.

[4] S.-T. Yuan and J. Sun, "Ontology-based structured cosine similarity in speech document summarization," in *IEEE/WIC/ACM International Conference on Web Intelligence (WI'04)*. IEEE, 2004, pp. 508–513.

[5] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: An asr corpus based on public domain audio books," in *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2015, pp. 5206–5210.

[6] P. B. . D. Weenink, "Praat: doing phonetics by computer [computer program]," Glot International, p. 341–345, (1992–2022).

[7] L. Moro-Velazquez, J. A. Gomez-Garcia, J. D. Arias-Londoño, N. Dehak, and J. I. Godino-Llorente, "Advances in parkinson's disease detection and assessment using voice and speech: A review of the articulatory and phonatory aspects," *Biomedical Signal Processing and Control*, vol. 66, p. 102418, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S174680942100015X

[8] B. M. Ansel and R. D. Kent, "Acoustic-phonetic contrasts and intelligibility in the dysarthria associated with mixed cerebral palsy," *Journal of Speech, Language, and Hearing Research*, vol. 35, no. 2, pp. 296–308, 1992.

[9] K. Rosen and S. Yampolsky, "Automatic speech recognition and a review of its functioning with dysarthric speech," *Augmentative and Alternative Communication*, vol. 16, no. 1, pp. 48–60, 2000. [Online]. Available: https://doi.org/10.1080/07434610012331278904

[10] R. Takashima, T. Takiguchi, and Y. Ariki, "Two-step acoustic model adaptation for dysarthric speech recognition," in *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 6104–6108.

[11] J. Shor, D. Emanuel, O. Lang, O. Tuval, M. P. Brenner, J. Cattiau, F. Vieira, M. McNally, T. Charbonneau, M. Nollstadt, A. Hassidim, and Y. Matias, "Personalizing ASR for dysarthric and accented speech with limited data," *CoRR*, vol. abs/1907.13511, 2019. [Online]. Available: http://arxiv.org/abs/1907.13511

[12] D. Mulfari, A. Celesti, and M. Villari, "Exploring ai-based speaker dependent methods in dysarthric speech recognition," in *2022 22nd IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2022, pp. 958–964.

[13] K. T. Mengistu and F. Rudzicz, "Comparing humans and automatic speech recognition systems in recognizing dysarthric speech," in *Advances in Artificial Intelligence*, C. Butz and P. Lingras, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011, pp. 291–300.

[14] D. Mulfari, G. Meoni, M. Marini, and L. Fanucci, "Machine learning assistive application for users with speech disorders," *Applied Soft Computing*, vol. 103, p. 107147, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1568494621000703

[15] K. L. Lansford, J. M. Liss, and R. E. Norton, "Free-classification of perceptually similar speakers with dysarthria," *Journal of Speech, Language, and Hearing Research*, vol. 57, no. 6, pp. 2051–2064, 2014.